

StackIT vs AWS

Deep Comparative Analysis

GDPR-Compliant Hybrid Storage Architecture

Bayer Pharma Revenue Management Pipeline

ModelN.io on AWS + EU Data Residency

Prepared by: Nova | March 2026

Executive Summary

Yes — running compute on AWS with storage on StackIT is **architecturally sound**.

Component	Where	Why
Compute	AWS	Glue ETL, Lambda, EC2 — no GDPR issue
EU PII Storage	StackIT EU01	100% German jurisdiction, no CLOUD Act
Network	HTTPS/TLS 1.3	Public internet, ~10-100ms depending on route

“ **Verdict:** Strongly conditional YES. Best for storage workloads where cross-cloud latency is acceptable.

”

What is StackIT?

Attribute	Details
Parent	Schwarz Group (Lidl + Kaufland, ~€140B revenue)
Division	Schwarz Digits (IT/Digital arm)
HQ	Neckarsulm, Germany
Founded	2018 internally, public ~2022
Positioning	"European Hyperscaler" — digital sovereignty
Customers	Sana Clinics, Schwarz Group internal

Two regions: EU01 (Germany), EU02 (Austria) — 3 AZs each

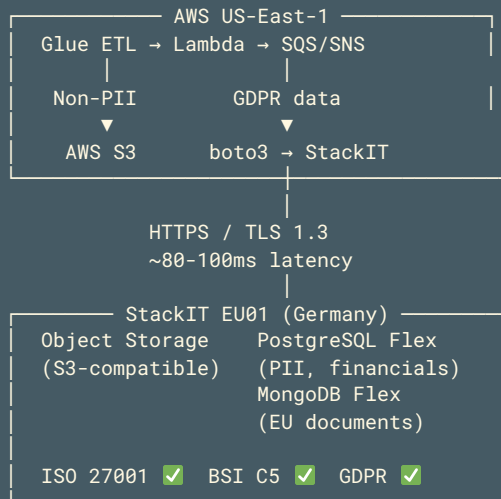
Data governance: 100% EU-based — data never leaves Germany/Austria

GDPR & Security Certifications

Certification	StackIT	AWS
GDPR (Art. 28 DPA)	✓ Fully compliant	✓ Available
ISO 27001	✓ TUV SUD certified	✓ Certified
BSI C5 Type 2	✓ Broad coverage	✓ Available
No CLOUD Act exposure	✓ German company	⚠ US-domiciled
Schrems II compliant	✓ No 3rd-country transfers	⚠ Requires SCCs/BCRs
Data encrypted at rest	✓ AES-256	✓ AES-256
Customer-managed keys	✓ SSE-C	✓ KMS + SSE-C

“ **Key advantage:** StackIT = German company, no US parent = **zero CLOUD Act exposure** ”

Architecture: AWS Compute + StackIT Storage



Latency Considerations

Route	Latency	Suitable For
AWS eu-central-1 → StackIT EU01	8-15ms	Real-time queries OK
AWS eu-west-1 → StackIT EU01	20-30ms	Async processing
AWS us-east-1 → StackIT EU01	80-100ms	Batch ETL only
StackIT AZ-to-AZ	0.5ms	Internal workloads

“ **Critical for Bayer:** ModelN.io runs on US-East-1. Expect 80-100ms to StackIT.

Fine for: Batch ETL, object storage writes, async processing

Not suitable for: Synchronous OLTP query patterns

”

Network Connectivity

Option	Status	Details
Public Internet (HTTPS)	✓ Available	Default, TLS 1.3, works today
AWS Direct Connect	✗ Not available	No StackIT-AWS dedicated link
Site-to-Site VPN	⚠ Possible	Self-managed on both ends
DE-CIX Peering	✓ EU02 only	Requires colocation

Today's reality: All traffic over public internet (encrypted).
No AWS Direct Connect equivalent exists for StackIT.

“ This is the **biggest architectural risk** — no private peering. ”

Service Comparison (Highlights)

Category	AWS	StackIT	Maturity
Object Storage	S3	S3-compatible	★★★★★
PostgreSQL	RDS/Aurora	PostgreSQL Flex	★★★★★★
MongoDB	DocumentDB	MongoDB Flex	★★★★★
Kubernetes	EKS	SKE	★★★★★
Redis	ElastiCache	Redis	★★★★★
Secrets	Secrets Manager	Secrets Manager	★★★★★
Serverless	Lambda	✗ N/A	—
Kafka	MSK	✗ (RabbitMQ only)	—
AI/ML	SageMaker	Early stage	★★

Pricing Comparison

Compute — StackIT is 15-60% Cheaper

Instance		AWS (eu-central-1)	StackIT EU01	Savings
1 vCPU / 4GB		~\$70/mo	€27/mo	~60%
2 vCPU / 8GB		~\$96/mo	€55/mo	~43%
4 vCPU / 16GB		~\$140/mo	€109/mo	~22%
8 vCPU / 32GB		~\$280/mo	€218/mo	~22%

Storage — Comparable

Metric		AWS S3	StackIT	Notes
Per GB/month		\$0.023	€0.027	Slightly higher
Cold tier		\$0.004 (Glacier)	€0.027	No cold tier yet

Managed Database Pricing

Service	AWS Price/Month	StackIT Price/Month	Notes
PostgreSQL 2CPU/4GB HA	~\$100	€142 (3 nodes)	StackIT: 3 replicas vs AWS 2
PostgreSQL 4CPU/8GB HA	~\$180	€272 (3 replicas)	Higher but more redundant
PostgreSQL Single	~\$50	~€70	Comparable

“ **Note:** StackIT defaults to **3-node replica clusters** for HA.
AWS Multi-AZ = 2 nodes. Not apples-to-apples.

”

Kubernetes

- AWS EKS control plane: **\$72/mo**
- StackIT SKE: **Included in worker pricing** (free control plane)

Terraform Provider Comparison

Dimension	AWS	StackIT
Registry	<code>hashicorp/aws</code>	<code>stackitcloud/stackit</code>
Tier	Official	Community
Version	~5.x (stable)	0.88.0 (pre-1.0)
Resources	~1,100+	~80-100
Downloads	4+ billion	6.49 million
Release Frequency	Weekly	Weekly

“ **⚠ Pre-1.0 Warning:** Breaking changes possible between versions.

Always pin: `version = "= 0.88.0"`

”

boto3 — Zero Code Changes

```
import boto3
from botocore.config import Config

# Just change endpoint_url - everything else works
stackit_s3 = boto3.client("s3",
    endpoint_url="https://object.storage.eu01.onstackit.cloud",
    aws_access_key_id="YOUR_STACKIT_KEY",
    aws_secret_access_key="YOUR_STACKIT_SECRET",
    config=Config(signature_version="s3v4",
        region_name="eu01")
)

# All standard S3 operations work:
stackit_s3.put_object(Bucket="eu-pii", Key="data.parquet", Body=data)
stackit_s3.get_object(Bucket="eu-pii", Key="data.parquet")
```

“ **Recommended:** Store StackIT creds in AWS Secrets Manager, inject at runtime.

”

Data Segmentation Strategy

Stays on AWS (no EU residency required)

- Raw vendor data before normalization (non-EU)
- Processing state / intermediate results
- AWS Glue catalog metadata
- Lambda function code and configs
- Non-EU customer data

Moves to StackIT EU01 (GDPR required)

- EU patient/customer PII
- Medicaid EU claims data
- European chargeback records
- Price lists with EU customer data
- Any data subject to GDPR Article 44+ restrictions

For Bayer Pharma Specifically

Data Type	Recommendation	Why
Patient PII	StackIT	EU residency guaranteed
PHI (Protected Health)	StackIT	ISO 27001 + BSI C5 = healthcare-ready
Financial records	StackIT	German jurisdiction, no US legal exposure
Medicaid EU claims	StackIT	GDPR Article 28 DPA compliant
US chargeback data	AWS	No EU residency requirement
ETL compute	AWS	Processing in flight, no GDPR issue
ML/Analytics	AWS	SageMaker + Glue more mature

Operational Complexity Added

Concern	Impact	Mitigation
Two cloud portals	Low	Document runbooks
Two billing accounts	Low	Cost allocation tags
Two credential systems	Medium	Centralize in AWS Secrets Manager
Two monitoring systems	Medium	Consolidate dashboards
Network dependency	Medium	Async patterns, retry logic
Terraform state	Low	Single state manages both 

Risk Assessment

Risk	Severity	Mitigation
Cross-cloud latency	Medium	Use async patterns, avoid sync US → EU queries
No direct peering	Medium	Accept latency or colocation via DE-CIX
Provider pre-1.0 changes	Medium	Pin Terraform provider version
StackIT service gaps	Low	Keep Lambda/Kafka on AWS
Egress costs	Medium	Measure volume, negotiate contract
Company risk	Low	Backed by €140B Schwarz Group

Recommendation

YES — Use StackIT for EU-Regulated Storage

Best for:

- EU PII object storage (S3-compatible)
- PostgreSQL Flex (structured EU data + GDPR)
- MongoDB Flex (EU document data)
- Archiving (audit-proof, long-term retention)

Keep on AWS:

- All compute (Glue, Lambda, EC2)
- Non-EU data
- Latency-sensitive OLTP
- Serverless / Kafka workloads

Next Steps

1. **Create StackIT account** → Request EU01 project
2. **Provision Object Storage** → Create PII bucket via Terraform
3. **Configure boto3** → Point `endpoint_url` to StackIT
4. **Migrate EU data** → `rsync copy` from AWS S3 to StackIT
5. **Set up PostgreSQL Flex** → 3-replica HA for structured EU data
6. **Store creds in AWS Secrets Manager** → Single source of truth
7. **Update Glue ETL jobs** → Route EU-sensitive writes to StackIT
8. **Monitor latency** → Track cross-cloud performance

Thank You

StackIT vs AWS — Hybrid Architecture
GDPR-Compliant Storage for Bayer Pharma

Nova | StrategyBase | March 2026

Full analysis: [STACKIT-VS-AWS-ANALYSIS.md](#)